Meta-CoT Control-V5: From Metacognitive Detection to Execution

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Abstract

The Control-V5 experiment series trains Qwen3-8B to emit structured metacognitive control signals (`<|meta|>` blocks) and evaluates whether these signals translate into measurable OOD test-time adaptation gains. Eleven model variants were evaluated on a pilot set of 90 problems. Meta-CoT signals are parseable and trainable via reward shaping, but the current rewards do not produce functional test-time controllers. The bottleneck has shifted from meta detection to meta execution: models diagnose weaknesses but fail to act on them. Three targeted fixes are proposed for the next round.

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1 Executive Summary

Table 1: Key model comparison (n=90 pilot, 30 per benchmark)

Model	Acc	ECE	Conf_Cov	Wrong_Hi	Key Observation
base_sft	42.2	n/a	0.0	n/a	Accuracy baseline; no meta
E9	41.1	0.301	91.1	50.9	Highest meta acc; verify decorative
E8	38.9	0.322	23.3	3.6	Overconf suppression works
E5	40.0	0.790	1.1	0.0	Reward escape via omission

Conclusion. Meta-CoT signals are parseable and trainable via reward shaping, but the current generation of rewards does not produce functional test-time controllers. The primary bottleneck has shifted from *meta detection* to *meta execution*: models diagnose weaknesses but fail to act on them. Three targeted fixes—same-route repetition penalty, route-switch evidence reward, and confidence omission floor—are proposed for the next round.

2 Background and Intent

2.1 Definition: Meta-CoT

Meta-CoT is a structured reasoning protocol in which a language model interleaves standard chain-of-thought computation with explicit metacognitive control blocks delimited by `<|meta|>` and `<|/meta|>` tokens. Each meta block contains three fields:

- **confidence**: a scalar in $[0, 1]$ representing the model's posterior belief in the current solution route.
- **assessment**: a natural-language diagnosis of why the current route may be incorrect.
- **action**: one of {verify, redirect, diagnose}, specifying the next controller step.

The term **controller** refers to the decision function that maps the current reasoning state to one of these actions. A functional controller would (a) detect when confidence is unjustified, (b) invoke verify with an independent check rather than a paraphrase, and (c) execute redirect by switching the solution route when diagnosis indicates a dead end.

2.2 Research Questions

- **RQ1 (Parseability and Adaptation):** Can Meta-CoT teach a parseable controller state that enables OOD test-time adaptation?
- **RQ2 (Learnability via Rewards):** Can Meta-RL make verify, redirect, and diagnosis behaviors learnable through reward decomposition?
- **RQ3 (Curriculum Extensibility):** Can curriculum/RAG extend from reliable diagnosis to improved performance?

2.3 Experimental Context

All models are fine-tuned from a Qwen3-8B base. SFT variants use supervised fine-tuning on 4,996 Meta-CoT chains generated by GPT-5.4-mini. RL variants apply GRPO with decomposed reward functions. Evaluation uses `max_tokens=4096`.

3 Experiment Design

3.1 Reward Decomposition Contract

The RL experiments form a progression from broad to targeted rewards:

E3: Full Meta + Correctness + Calibration. **Intent:** Combine all reward components. **Hypothesis:** Joint optimization will produce an emergent controller. **Verification:** Accuracy $\geq$ base_sft; ECE < 0.3; Wrong_Hi $< 20\%$.

E5: Correctness-Only RL. **Intent:** Remove all meta rewards. **Hypothesis:** Without meta rewards, no meta blocks emitted; accuracy preserved. **Verification:** Accuracy $\geq$ base_sft; Conf_Cov ≈ 0.

E8: Overconfidence Penalty + Correctness. **Intent:** Penalize high-confidence wrong answers. **Hypothesis:** Wrong_Hi will decrease without destroying accuracy. **Verification:** Wrong_Hi $< 10\%$; accuracy within 3%p of base_sft.

E9: Verify Reward + Correctness. **Intent:** Reward verify actions on uncertain completions. **Hypothesis:** Verify frequency and accuracy will improve. **Verification:** Conf_Cov $> 80\%$; AIME improvement.

E10: Full Controller (Verify + Redirect + Diagnosis). **Intent:** Train the complete metacognitive controller. **Hypothesis:** Full reward stack produces detect–diagnose–recover behavior. **Verification:** Accuracy $\geq$ base_sft; diagnosis-to-recovery $> 50\%$.

3.2 SFT Ablations

Model	Training Data
all_sft	Full Meta-CoT with verify + redirect + diagnosis
verify_sft	Chains containing only verify meta blocks
redirect_sft	Chains containing only redirect meta blocks

4 Quantitative Results

4.1 Control-V5 Pilot (n=90)

Table 2: Full pilot results (30 problems per benchmark)

Model	Acc	AIME	GSM8K	MATH500	Conf_Cov	ECE	Wrong_Hi
base_sft	42.2	13.3	80.0	33.3	0.0	n/a	n/a
all_sft	33.3	6.7	63.3	30.0	88.9	0.398	51.7
verify_sft	36.7	3.3	66.7	40.0	82.2	0.492	80.7
redirect_sft	35.6	10.0	63.3	33.3	48.9	0.160	0.0
E3	30.0	3.3	63.3	23.3	100.0	0.515	71.4
E5	40.0	3.3	80.0	36.7	1.1	0.790	0.0
E8	38.9	10.0	80.0	26.7	23.3	0.322	3.6
E9	41.1	6.7	90.0	26.7	91.1	0.301	50.9
E9b	40.0	6.7	83.3	30.0	0.0	n/a	n/a
E9c	36.7	6.7	73.3	30.0	7.8	0.264	1.8
E10	35.6	6.7	66.7	33.3	5.6	0.340	0.0

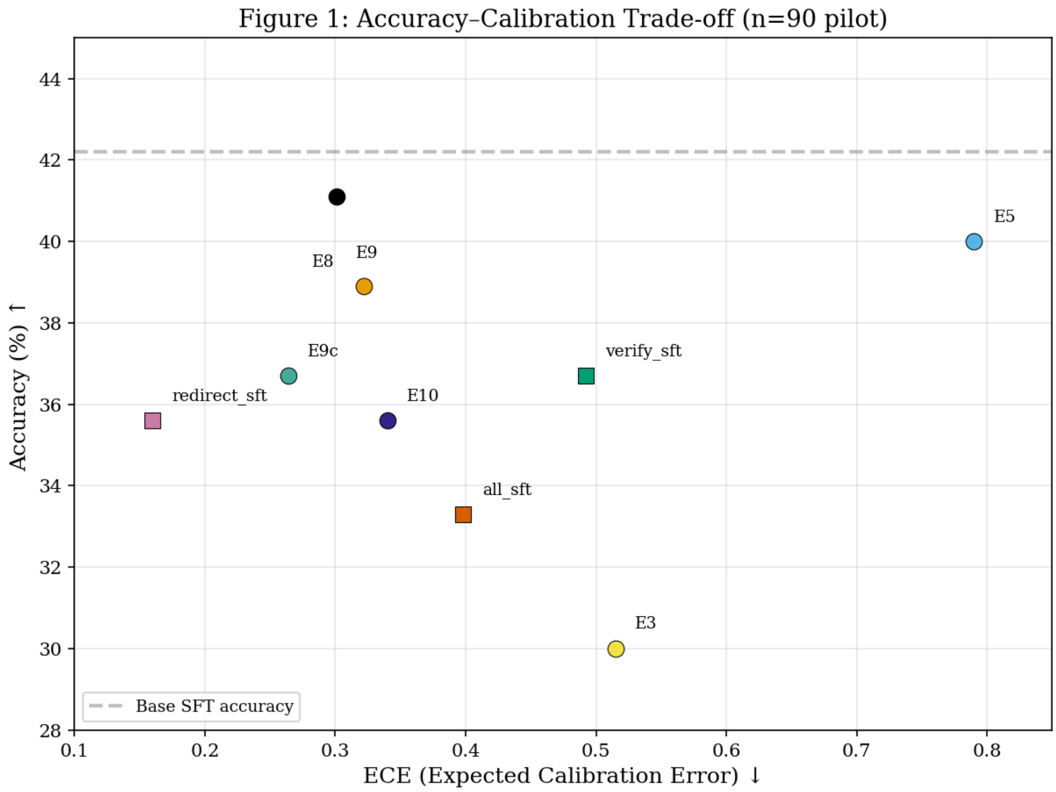

Figure 1: Accuracy–Calibration trade-off. Models with ECE defined (Conf_Cov > 0) plotted. Dashed line = base_sft accuracy.

Key observations:

1. No meta model exceeds base_sft accuracy (42.2). The highest meta model is E9 at 41.1.
2. E8 achieves Wrong_Hi < 10% (3.6%) while maintaining accuracy within 3.3%p of base_sft.
3. E3 shows the worst accuracy (30.0) despite 100% confidence coverage.
4. E5, E9b, and E10 have confidence coverage below 10%, rendering calibration metrics unreliable.

4.2 Per-Benchmark Analysis

E9 achieves 90.0 on GSM8K (highest in pilot), indicating verify rewards can improve arithmetic self-checking. However, E9 scores only 26.7 on MATH-500, suggesting verify does not transfer to harder reasoning. AIME is the hardest benchmark, and meta overhead actively harms performance: only base_sft achieves 13.3.

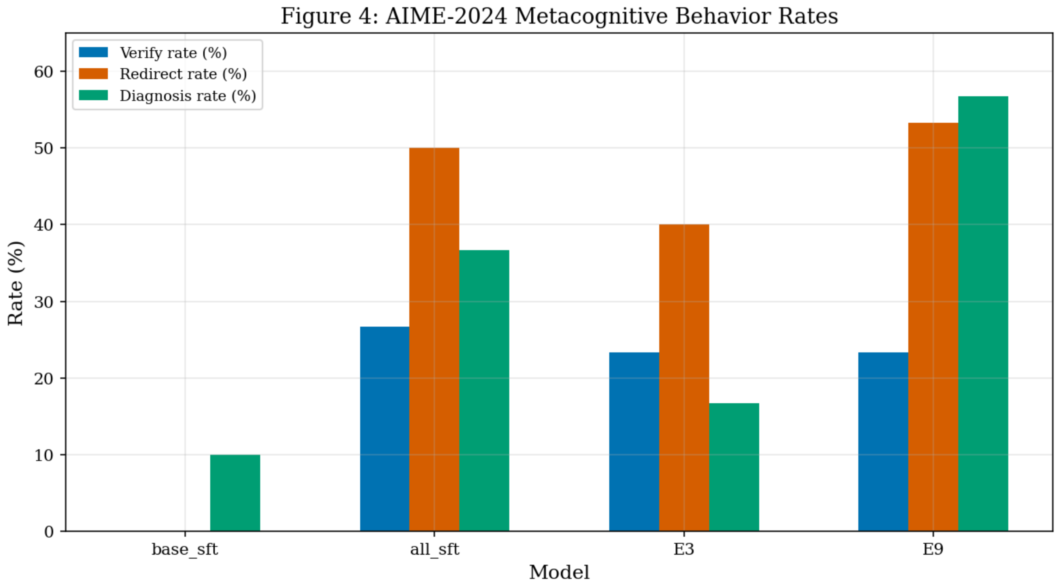

Figure 2: AIME-2024 metacognitive behavior rates across selected models.

4.3 Large-Scale Reference (1,030 problems)

Table 3: Previous-round 1,030-problem evaluation

Model	Overall Accuracy
V2 SFT	72.72%
V3 SFT	72.0%
E5	72.04%
Base SFT	71.7%
E8	67.2%

Figure 3: Multi-panel summary: (A) per-benchmark accuracy, (B) overall ranking, (C) ECE, (D) confidence coverage.

5 Qualitative Failure Analysis

Failure modes were identified by manual inspection of wrong-answer completions (source: `failure_analysis_2026_04_04.json`).

5.1 Overconfident Verify Failed

Affected models: E3 (45 cases), E9 (15 cases), verify_sft.

The model completes an initial solution, emits a meta block stating "single route, should verify," then re-derives the same calculation using identical assumptions. Confidence remains high (0.86–0.89). The verification step produces a *paraphrase* of the original reasoning rather than an independent falsification.

Example (E3, GSM8K, "sheep counting"): The model computes total $= 160 + 80 + 20 = 320$. Meta block: "confidence: 0.88. I should do a brief independent check." Verification: recalculates $20 + 80 + 160 = 320$. Answer remains 320 (wrong; correct $= 260$).

Root cause: The verify reward triggers on verification *language*, not verification *independence*.

5.2 Diagnosis Without Recovery

Affected models: E8 (4 cases), E10 (1 case), E3 (4 cases).

The meta block correctly identifies the weakness and specifies a `study_need`. However, the subsequent reasoning continues along the original route without executing the switch.

Example (E10, AIME "Aimeville sets"): Meta diagnoses need for inclusion-exclusion. Subsequent reasoning: "437 + 234 + x = 900, x = 229." No inclusion-exclusion attempted. (Correct answer: 73.)

Root cause: The redirect reward credits diagnosis text without requiring execution evidence.

5.3 Single Intervention Only

Affected: all_sft, E8, E10. A single meta block appears early; no further monitoring follows, even on hard problems.

Root cause: Reward functions do not incentivize repeated intervention.

5.4 Reward Escape via Confidence Omission

Affected: E5 (54 cases), E10 (53 cases). The model omits meta blocks entirely, avoiding calibration penalties.

Root cause: The calibration reward returns 0.0 (neutral) for missing confidence, making omission cost-free.

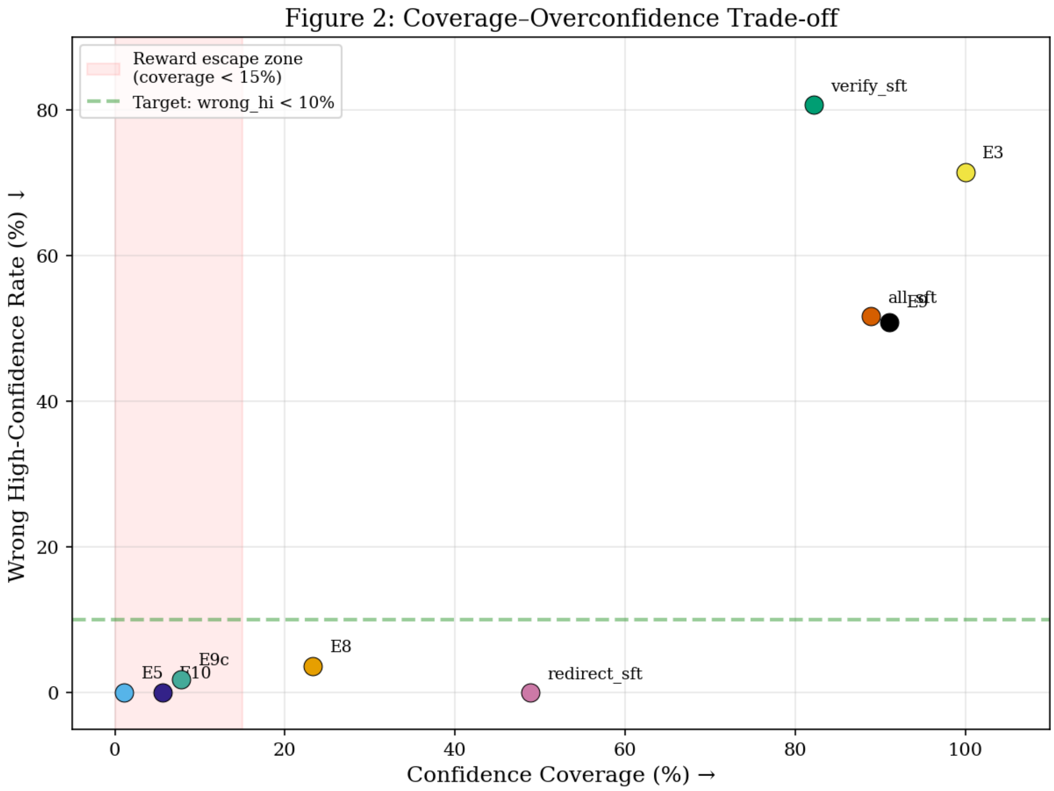

Figure 4: Confidence coverage vs. wrong high-confidence rate. Red zone indicates reward escape (coverage $< 15\%$).

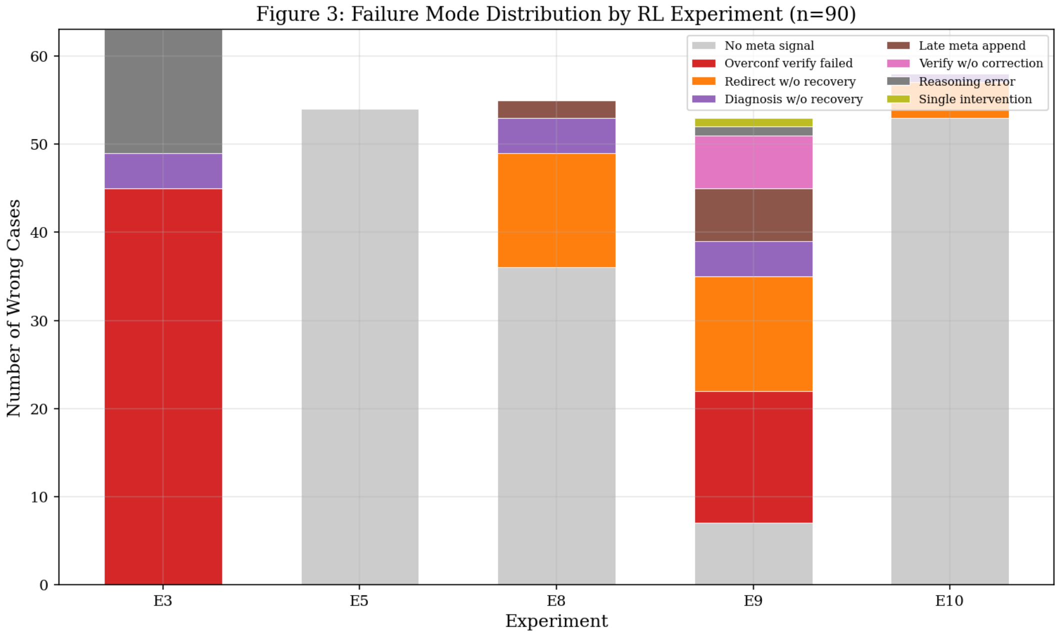

Figure 5: Failure mode distribution across RL experiments (n=90 pilot).

6 Hypothesis Evaluation

6.1 RQ1: Parseable Controller State for OOD Adaptation

Verdict: Weak support. Meta blocks are parseable across all meta-emitting models. However, no meta model exceeds base_sft accuracy (42.2 vs. 41.1). OOD adaptation is not demonstrated. The controller state is syntactically present but semantically inert.

6.2 RQ2: Learnability via Reward Decomposition

Verdict: Partial support.

- **Verify (partial):** E9 achieves 90.0 on GSM8K but Wrong_Hi = 50.9. Verify is decorative.
- **Overconfidence suppression (supported):** E8 reduces Wrong_Hi to 3.6.
- **Redirect/diagnosis (not supported):** E10 accuracy = 35.6; 53/58 wrong answers have no meta signal.

6.3 RQ3: Curriculum Extensibility

Verdict: Gated. Diagnosis quality is insufficient for curriculum. Study_need fields are topically correct but not acted upon. Curriculum gating is maintained until diagnosis-to-recovery > 50%.

7 Proposed Improvements

7.1 Fix 1: Same-Route Repetition Penalty (Verify Quality)

Problem: 45/63 wrong answers in E3 and 15/53 in E9 exhibit same-route verify repetition.

Solution: Add a penalty (−0.5) when the solve tail after a verify meta block contains repetition language without independent method signals.

Expected effect: Shift verify from paraphrase to independent cross-check; reduce overconfident_verify_failed cases.

7.2 Fix 2: Route-Switch Evidence Reward (Redirect Execution)

Problem: Redirect meta blocks produce correct diagnoses but subsequent reasoning continues on the original route.

Solution: Add a reward (+0.9 correct, +0.25 wrong) when the post-redirect solve tail shows structural method difference from the prefix; penalty (-0.3) when switch is announced but not evident.

Expected effect: Convert diagnosis-without-recovery to functional redirects.

7.3 Fix 3: Confidence Omission Floor (Prevent Reward Escape)

Problem: E5 (1.1% coverage) and E10 (5.6%) minimize calibration loss by omitting confidence entirely.

Solution: Penalty (-0.5) per completion with zero meta blocks.

Expected effect: Force confidence emission; enable calibration metrics to operate across the full distribution.

8 Next Experiments

8.1 E9v2: Verify with Repetition Penalty

Applies Fix 1 to E9. **Hypothesis:** Wrong_Hi will decrease from 50.9 while GSM8K accuracy stays $\geq$ 85%. **Success criterion:** Wrong_Hi $<$ 30%; GSM8K $\geq$ 85%.

8.2 E9bv2: Redirect with Route-Switch Evidence

Applies Fix 2 to E9b. **Hypothesis:** Diagnosis-with-recovery rate will exceed 50%. **Success criterion:** Conf_Cov $\geq$ 30%; accuracy $\geq$ 38%.

8.3 E10v2: Full Controller with All Fixes

Applies all three fixes to E10. **Success criterion:** Accuracy $\geq$ 42.2 (base_sft parity); $\geq$ 2 of 4 failure modes reduced by $\geq$ 50%.

8.4 Full-Scale Evaluation Plan

Models meeting pilot criteria will proceed to the 1,030-problem evaluation (GSM8K 500, MATH-500 500, AIME 30) with `max_tokens=4096` and deterministic decoding.

All numerical results drawn from the Control-V5 pilot evaluation (n=90, 30 per benchmark) unless otherwise noted. Source: `control_v5_eval_readout_2026_04_04.md`